Classifying nonstationary time-varying signals on graphs through their deviation from a stationary behavior: preliminary results on EEG signals

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Abstract

We consider here the classification problem of finding the Motor Imagery (MI) class. As far as classical approaches do not consider the structure of EEG signals, newer approaches based on graph signal processing have been developed.

While many of these techniques assume stationarity and/or smoothness of graph signals , which does not match the signal model considered here, In this work, we propose to discuss a possible regularization of nonstationarity, by studying the deviation of an EEG signal from its stationary behavior on a graph that we model by the covariance matrix of the electrodes. We are going to measure the within trial nonstationarity and improve the robustness of features trough filtering of EEGs. We show here our first observations on this methodology.

Background

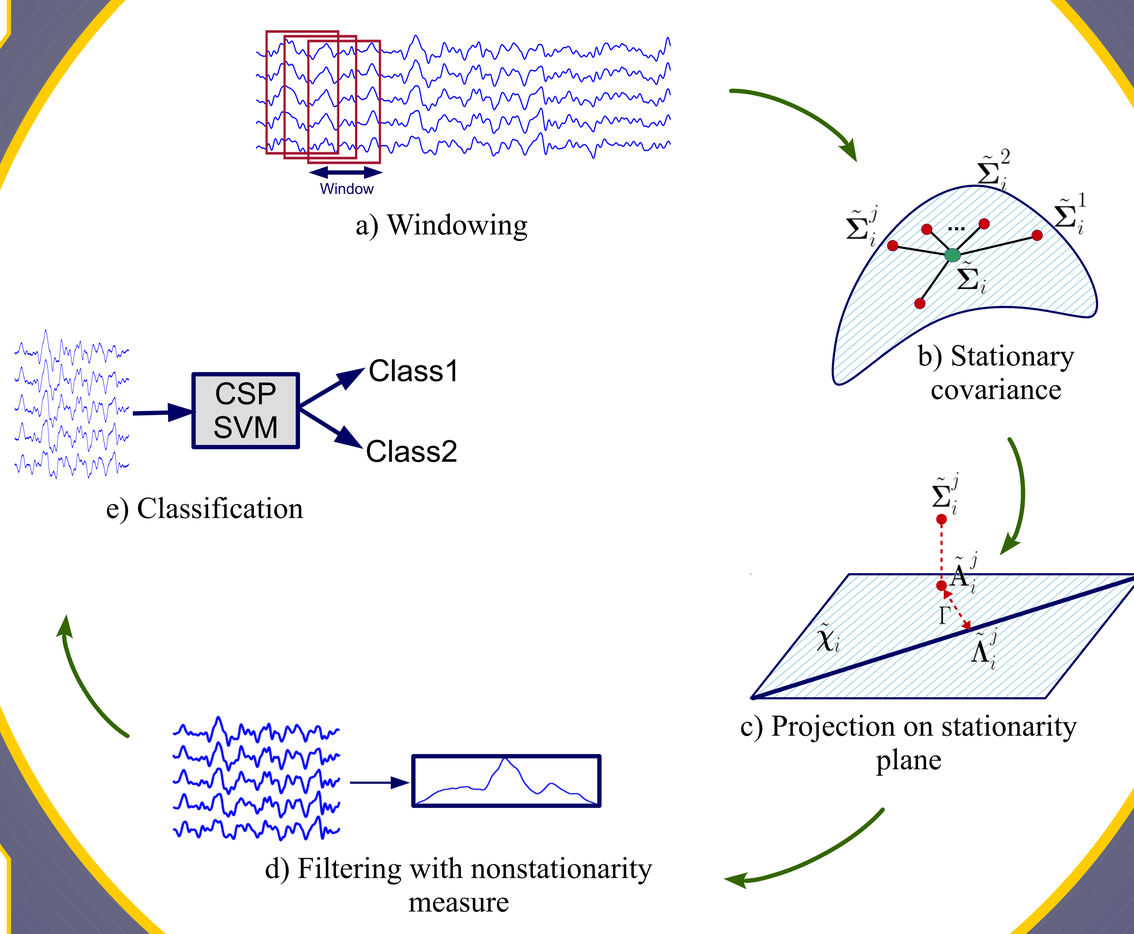
CSP:

The main idea of CSP is to find a linear transformation, that maximizes the variance of the one class while minimizing the variance of the other class at the same time.

$$w^* = \arg\min_w \frac{w^T \bar{\Sigma}_1 w}{w^T \left(\bar{\Sigma}_1 + \bar{\Sigma}_2\right) w}$$

Data set:

For evaluation of our method, we use BCI competition III data set IVa after frequency filtering in the range of [2Hz-40Hz] and time interval [0.5sec-2.5sec] after appearance of visual cues.



Method

a. We compute sample covariance matrices over windowed EEGs.

b. Afterward barycenter of sample covariances are calculated:

$$\tilde{\Sigma}_i = \arg\min_{\Sigma \in \mathcal{M}_{\text{spd}}} \sum_j d(\Sigma, \tilde{\Sigma}_i^j)$$

c. We compute the projections of sample covariances in the eigenbasis of barycenter covariance. And because the eigenvectors of sample covariances are not aligned with eigenvectors of barycenter, we approximate it with its diagonal elements:

$$\forall j : \tilde{\mathbf{A}}_i^j = \tilde{\boldsymbol{\chi}}_i^\top \tilde{\boldsymbol{\Sigma}}_i^j \tilde{\boldsymbol{\chi}}_i$$

At last we measure the nonstationarity:

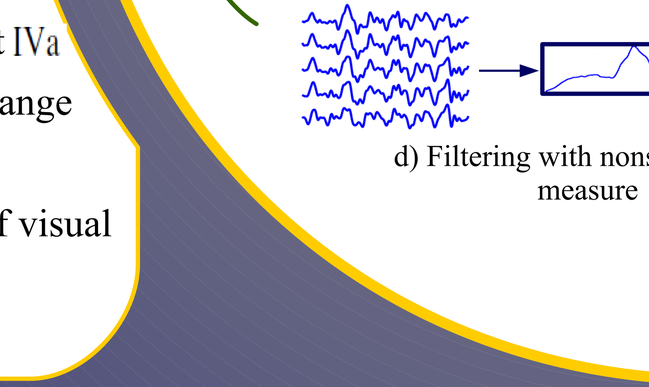
$$\Gamma(\tilde{\mathbf{A}}_i^j, \tilde{\boldsymbol{\Lambda}}_i^j) = \left\| \tilde{\mathbf{A}}_i^j (\tilde{\boldsymbol{\Lambda}}_i^j)^{-1} \right\|_{opt}$$

d. EEGs are filtered by this measure.

e. we classify filtered signals.

Results

Nonstationarity measure of one sample EEG signal and filtered EEG with this measure are shown in the following.
the peaks in the nonstationarity measure curve appear to match some patterns that appear in the EEG signals.



We classified BCI competition data set with this measure, mean and variance of error are shown in the following.

Table1 depicts that our method stabilizes the error of CSP method.

	CSP				Our Method			
	S1	S2	S3	S4	S1	S2	S3	S4
Mean	28.11	38.58	27.44	31.19	24.8	35.16	27.03	28.38
Var	0.0117	0.0064	0.0114	0.0165	0.0043	0.0056	0.019	0.0088

Table I: SVM Error

Future work

- How can we reach better accuracies?
- How to use this method for multiclass classification?
- Whats the effect of averaging metric?

References

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[3] Blankertz, et al. "The non-invasive Berlin brain–computer interface: fast acquisition of effective performance in untrained subjects." NeuroImage 37.2 (2007): 539-550.

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